
Assignment 3

Stochastic Simulation

Academic Year 2025–2026

University of Twente

In this assignment, you develop a simulation model that you will later use in the Project of Module 8 (202200339). Here, your assignment is to simulate the current situation at the CT scan department that is described in the Project, excluding the elements you are asked to provide advice on in the Project.

Deadline Notice

Final submission deadline for Assignment 3 is **22nd June 2025 at 23:59**. Submissions after this date will not be accepted.

Deadline: 22nd June 2025 at 23:59

Deliverables: Report (PDF, max. 15 pages) + Python code

General Instructions

General Instructions for All Assignments

You must complete the assignment described in this document according to the workflow below.

1. Read the base library repository.

The course provides a base Python library that implements core DES components. Before writing code, read and understand the repository carefully:

Repository GitHub: https://github.com/AE452/des_library

Repository GitLab: https://gitlab.utwente.nl/m7747128/des_library

Make sure you understand the class structure, event scheduling interface, and provided examples.

2. Fork the repository and set up Git.

Create your own fork and work there. Basic Git knowledge is **required**. At minimum, be comfortable with:

- `git clone <url>`
- `git add`, `git commit -m "..."`
- `git push`
- optional branching: `git branch`, `git checkout`

Reference: <https://git-scm.com/book/en/v2/Git-Basics-Getting-a-Git-Repository>

3. Implement Assignment 3 inside the fork.

Implement the CT scan department simulation in your fork. Your commit history should clearly reflect your development process.

4. Do *not* modify the base library code.

Build on top of the library interfaces. Recommended structure:

- `solution.py`: main simulation logic (classes, events, state, statistics).
- `notebook.ipynb` (optional): plots/tables only, importing from `solution.py`.

5. Submission

Submit on Canvas both a **PDF report** (max. 15 pages) and your **Python code**. Submit a link to a **public repository** hosted on GitLab or GitHub. If the repository is private, the submission is invalid.

The latest commit made **before the deadline** will be evaluated. The repository must contain:

- A fork of the original library
- A report (PDF, max. 15 pages)
- Python code (`solution.py` + optional notebook)

You may use the same repository as previous assignments. Store Assignment 3 deliverables in a folder named `Assignment3/`.

Important Warning

The code must run without any issues. I will not debug your code for you. If your code does not execute properly, it will be considered as not working.

6. Scoring and grading.

The total score is composed as follows.

Component	Weight
Analytical capacity check	10%
Event structure	35%
Data collection	20%
Implementation	20%
Results & interpretation	15%

Problem Description

Specifically, develop a simulation model for the situation described in the Project under ‘*Redesigning a CT scan department*’, where the current situation—which you are asked to simulate here—is as follows:

- The radiology department currently has **two CT scanners**. Both are open each weekday from 8:00 till 16:00, and one of the two stays available during evenings, nights, and weekends to scan emergency patients and inpatients whose scan requests arrive during those periods.
- Currently, the CT department does **not** use a dynamic blueprint.
- The **current policy for patient scheduling** is as follows: an outpatient gets an appointment as soon as possible (at the earliest on the day after the call), with the restriction that in the morning (from 8:00 till 12:00) only **4 slots** are available per hour and in the afternoon (from 12:00 till 16:00) only **3 slots** are available per hour. If there are no appointment slots available in the same week, the outpatient is put on a waiting list. On Friday evening, all outpatients on the waiting list are scheduled in appointment slots in the next week.
- Upon their arrival at the CT scan department, patients of all three patient types wait in the waiting room. **Emergency patients are treated with priority**; inpatients and outpatients are currently treated in order of arrival.
- Currently, there are **three chairs** in the waiting room. In this simulation study, ignore the PAS patients; so they do not occupy chairs in the waiting room.

Problems to be Considered

1. Analytical capacity check.

Examine whether two CT scanners are sufficient to be able to treat all patients.

Note: do *not* use simulation to answer this problem.

2. Discrete Event Simulation.

Examine the system using Discrete Event Simulation. In all cases: provide **95% confidence intervals** for the following performance indicators with **10% relative precision**.

- **CT scanner utilization**, with a distinction between office hours and outside office hours.

- **Access times for outpatients** (the number of days between the request and the actual appointment).
- **Waiting times** for emergency patients and for outpatients (the time between the patient's arrival in the waiting room and the start of the scan).
- **The fraction of (CT) patients** who have to wait outside the waiting room.
- For **inpatients with requests during office hours**, the percentage of those that cannot be scanned before 16:00 on the same weekday.

Final Report Structure

Here is a general structure of a report. An overall view gives that a report always consists of an introduction, a middle part in which the whole study is documented, and a conclusion. The introduction and the conclusion should be understandable for people who do not know mathematics nor simulation. The middle part should be understandable for fellow students.

In the following we give the sections and what they should contain. Note that the titles of the sections are an, albeit very short, abstract of the section. Also the title page is the first invitation for people to read your report, so think of a good title.

Introduction

Start with a general, high-level view of the situation. Continue with what is already known about this problem and how you will find the solution. As last, present an overview of your report. Keep in mind that not every reader will be interested in the complete report: some may be more interested in your approach to the problem, others may just want to know your solution. The introduction should be written with this diverse audience in mind. If the length of this section is too short, you can combine it with the following section *“Problem description in words”*.

Problem Description in Words

This section contains the full problem description in words. Describe the real-world situation, the problem that is encountered, and the performance measures that are studied in this report. This part should be written with the problem owners in mind; make clear that you are working on their problem.

Simulation Model

In this part you describe the assumptions and the events and give the pseudocode or flowchart for every event. Here you should also describe the way you will verify the simulation code.

Statistical Analysis

Here you describe the data that you collect during the simulation and how you use this data to find confidence intervals for the wanted performance measures.

Results

The big moment: present your results (always with CIs), both for the general case and the verification, and indicate to the reader which is interesting. Next to this, compare the analytical results for the special case with the data and, of course, the solution to the problem. Here you have to refer to your data to back up your conclusions.

Conclusion

Give the solution to the real-world problem in simple words understandable for a broad audience.

General Grading Criteria

Component	Description
Analytical capacity check	Correct reasoning without simulation.
Event structure	Correct identification of event types and interactions.
Data collection	Appropriate state variables and accumulators at each event.
Implementation	Correct, readable Python code reproducing required metrics.
Results & interpretation	Metrics reported with 95% CIs and meaningful discussion.

Grading Policy

Assignment 3 is graded with a numeric mark. The minimum required performance to pass is **50%**.

The use of AI/LLM tools for coding assistance is permitted; however, the submitted report—particularly the event descriptions and pseudo-code—must demonstrate your own understanding of discrete event simulation. All work must be written in your own words. Submissions that are identical or substantially similar to others will be treated as academic misconduct. Reliance on generated solutions without proper comprehension is not allowed. The use of AI tools is discouraged and should be limited to cases where you are genuinely stuck, serving as a learning aid rather than a source of direct answers. Any use of such tools must be clearly documented, including the prompts used (e.g., screenshots) and a brief explanation of how they contributed to your reasoning.

You must be able to clearly explain and justify every part of your submission if requested. This may include being asked to attend a tutorial session to further clarify your work.